

Exploring the Reasons Behind Land Cover Change and Evaluating the Management Strategies of a Selected Protected Area of Bangladesh

Background

❑ Declaration of forestlands as protected areas is considered as an important strategy to reduce deforestation and maintain ecological balance. At present, 30 percent of total tropical, subtropical and oceanic forest areas are legally declared protected areas (FAO, 2020). Regardless of all the conservational efforts, protected forest areas continue to degrade and lose forest cover.

❑ Forest cover constitutes for only 14.1% of the total land area in Bangladesh (FAO, 2020). Unfortunately, even this forestland cover has decreased at an alarming rate as a consequence of continuous deforestation. Because of ineptness of conventional management system to conserve the forest areas, the Bangladesh Government first initiated the process of declaring protected areas back in 1960 (BFD, 2024).

❑ Until now, 53 protected areas have been declared in Bangladesh, covering 469871.613 hectares land areas, representing 3.18% area of the country (BFD, 2024). National parks, eco-parks, game reserves and wildlife sanctuaries, and special biodiversity conservation zones are included in protected areas.

❑ But, destruction of forest areas has worsened over time in Bangladesh (Masum et al., 2023). For example, Bangladesh had 2.22 million hectares of natural forest in year 2010, covering 16% of total land area; where, 17.8 kilo hectares of natural forest land were lost in year 2023 (GFW, 2023).

Objective of the Study

1.To explore the reasons behind land cover change of a protected area in Bangladesh.

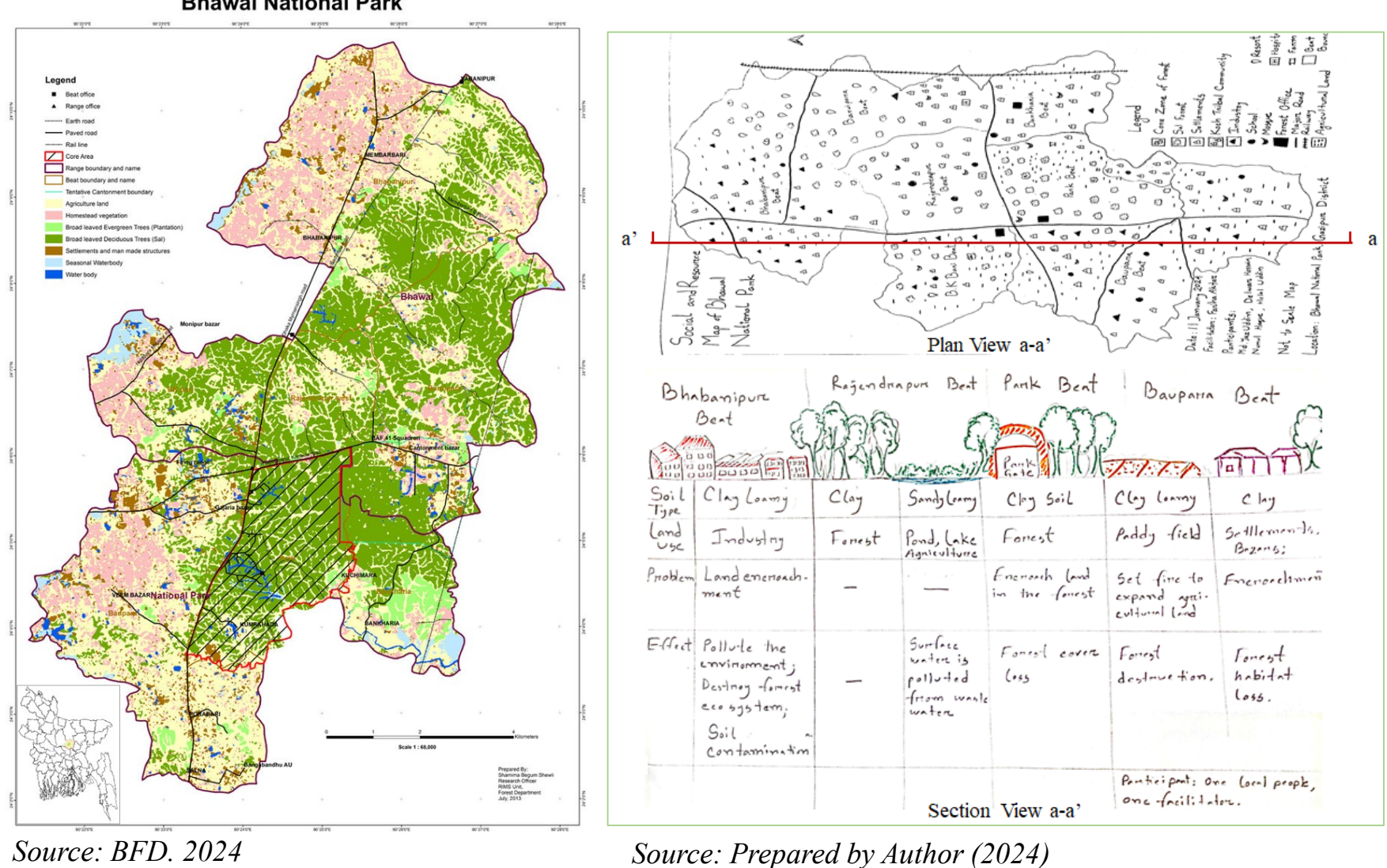
2. To evaluate management strategies for the selected protected area.

Study Area Selection

❑ In 1982, Bhawal forest was gazetted as a national park under the administrative control of Bangladesh Forest Department. 36% Sal forest cover existed in 1985, only 10% forest cover is remained, 0.12 million hecres of land (BFD, 2023). Bhawal Sal forest is one of the most threatened protected area because of increased human pressure on land (Ahsan et al., 2016).

❑ Bhawal National Park possess some distinctive characteristics.

- Connected with an important national highway (N3-Dhaka-Mymensingh Highway)
- Only at 40 kilometers distant from the capital Dhaka city.



Source: BFD, 2024

Source: Prepared by Author (2024)

Rationale of the Study

❑ Protected forestlands are under continuous threat worldwide due to population growth, rapid urbanization and increased demand for resources (Liu et al., 2023).

❑ Protected areas of Bangladesh are constantly converted into agricultural lands, settlements; making the country's conservation effort more challenging (Masum et al., 2023; Rahman et al., 2023). Previous studies in Bangladesh only concentrated on identifying land cover change in protected areas through quantitative method.

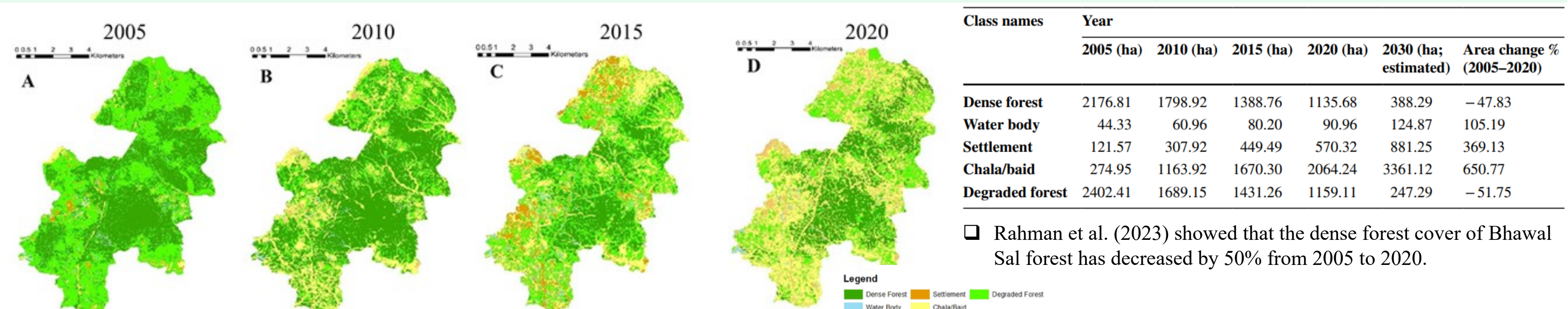
❑ Even though drivers of deforestation are generic worldwide such as population growth, infrastructural development, agriculture etc., understanding specific context and identifying the under lying causes are crucial to effectively manage the protected areas (Twongyirwe et al., 2018).

❑ Lack of effective management of protected areas have been a prominent cause of land conversion around the world (Hockings, 2003). Till date many management strategies were undertaken by BFD for conserving the areas, however the continuing deforestation question the existing management approach. For formulating adaptive strategies to address the threats and vulnerabilities of the protected areas, first it is important to assess the effectiveness of present management strategies through proper methods (Hockings, 2006).

Research Outcome

❑ This research will shed light on the internal and external threats to the selected protected area and provide necessary directions for adaptive management.

❑ The findings of this study will help the policymakers to comprehend the current status of protected area management system from diversified stakeholders' perspective and undertake required interventions to ensure effective management of protected areas.



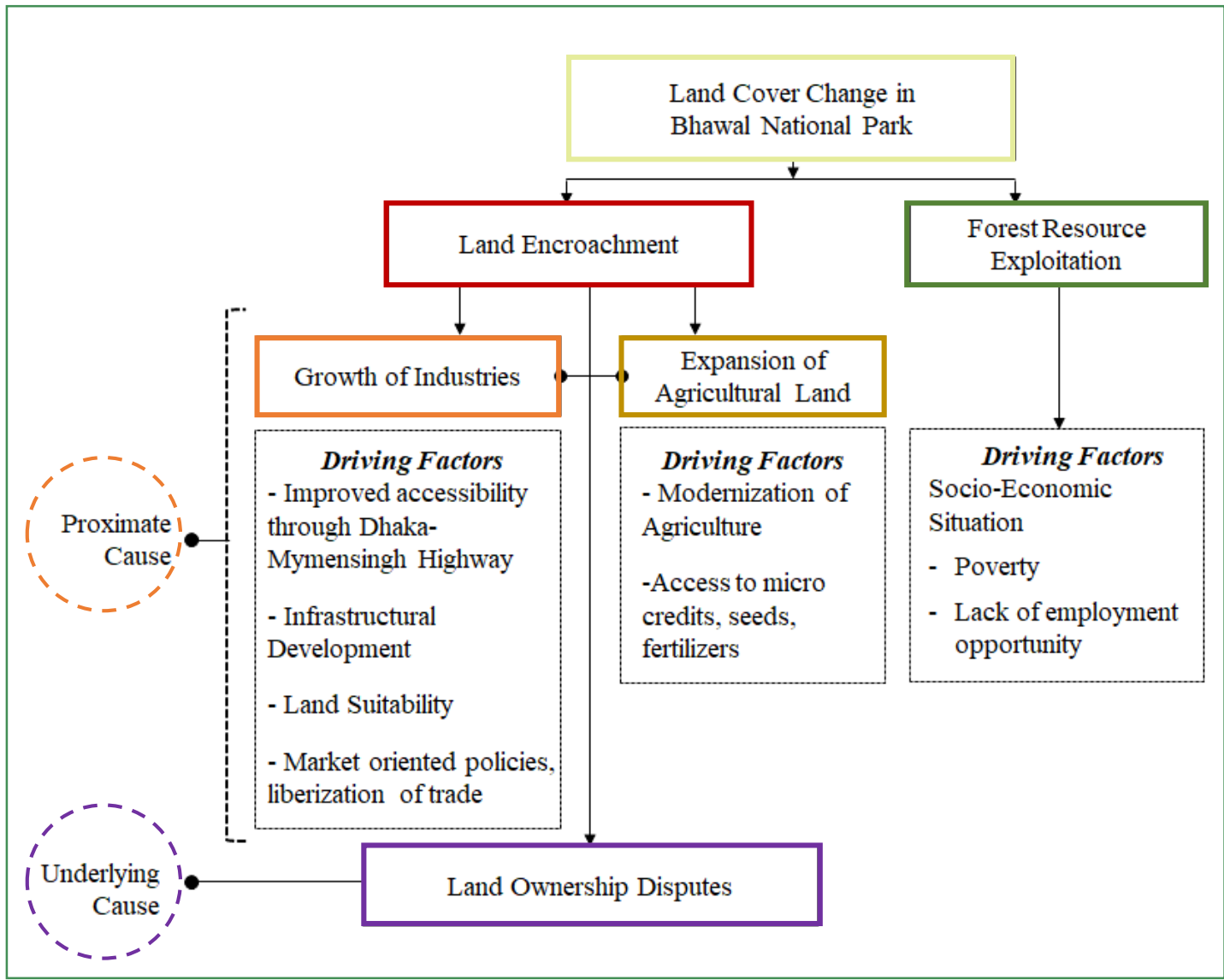
Temporal Change in the Study Area	
Timeline	Incidence
1950	Bhawal Zamindar ruled the area. Koch Tribal has been living for 100 years
1980	Dhaka-Mymensingh Highway (N3) Construction
1982	Declaration of Bhawal forest as a protected area. Sal forest restoration programme was undertaken.
1988	After the massive flood, industries started to shift in this area for having high elevation
1994	Establishment of primary school in Koch community
1995	There were 60-70 households in the indigenous community.
2000-2010	Development activities started, construction of road networks, electricity lines, The forest land cover was greatly destroyed during this period.
2024	Around 200 households in Koch community.

Source: Prepared by Author (2024) from KII and FGD

Identification of Causes behind Land Cover Change								
Causes of Land Cover Change	LE	ID	AG	LD	FR	IND	Frequency	Rank
Land encroachment (LE)		LE	LE	LD	LE	LE	3	II
Infrastructural development (ID)			AG	LD	ID	IND	1	V
Expansion of agricultural land (AG)				LD	AG	IND	2	IV
Land ownership disputes (LD)					LD	LD	5	I
Forest resource exploitation (FR)						IND	0	VI
Industrialization (IND)							4	III

Source: Prepared by Author (2024) from KII and FGD

Causes behind Land Cover Change of Bhawal National Park



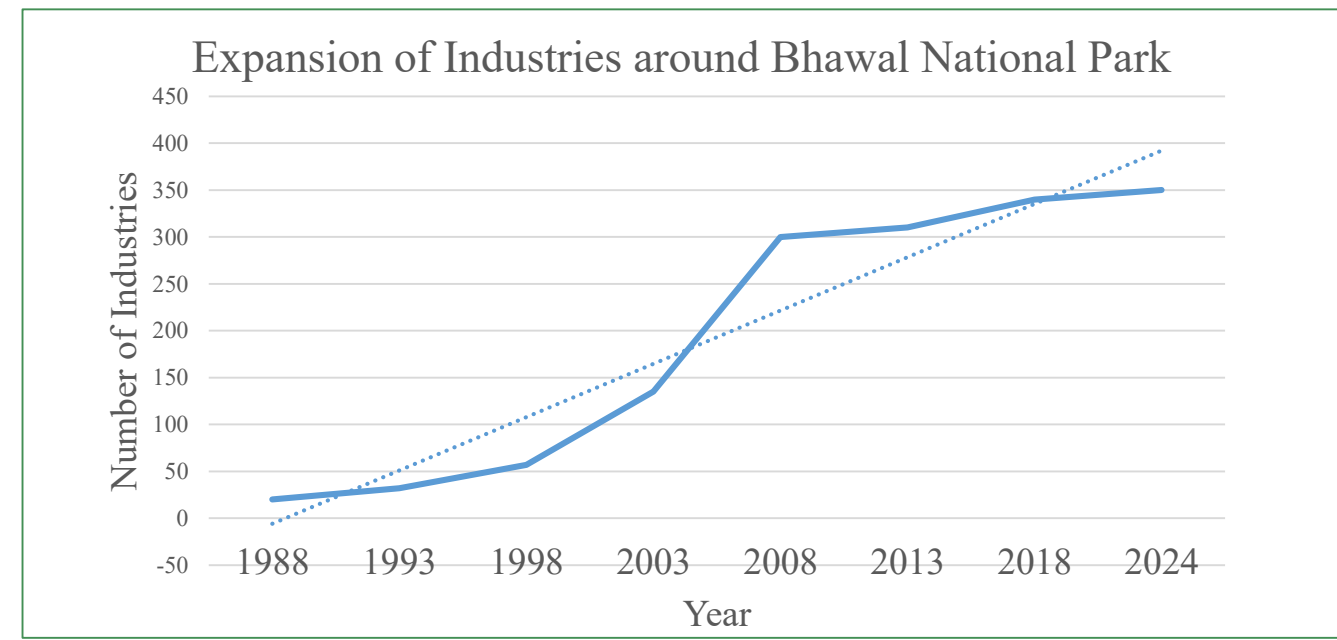
Source: Prepared by Author (2024) from KII and FGD

Land Ownership Disputes Regarding the Forestlands	
Timeline	Incidence
1940	Total lands of Bhawal was recorded through Cadastral Survey
Before 1950	Under the feudal system, Bhawal landlords possessed the lands.
1950	The East Bengal State Acquisition and Tenancy Act broke the feudal system. SA Khatiyon was prepared.
1950-1956	Court of Wards retained the lands
1956	Total recorded land in CS survey was entitled to BFD
1961-1965	Revisional survey (RS) was conducted, faulty recorded lands under individual names.
1999	Record correction cases was filed by BFD

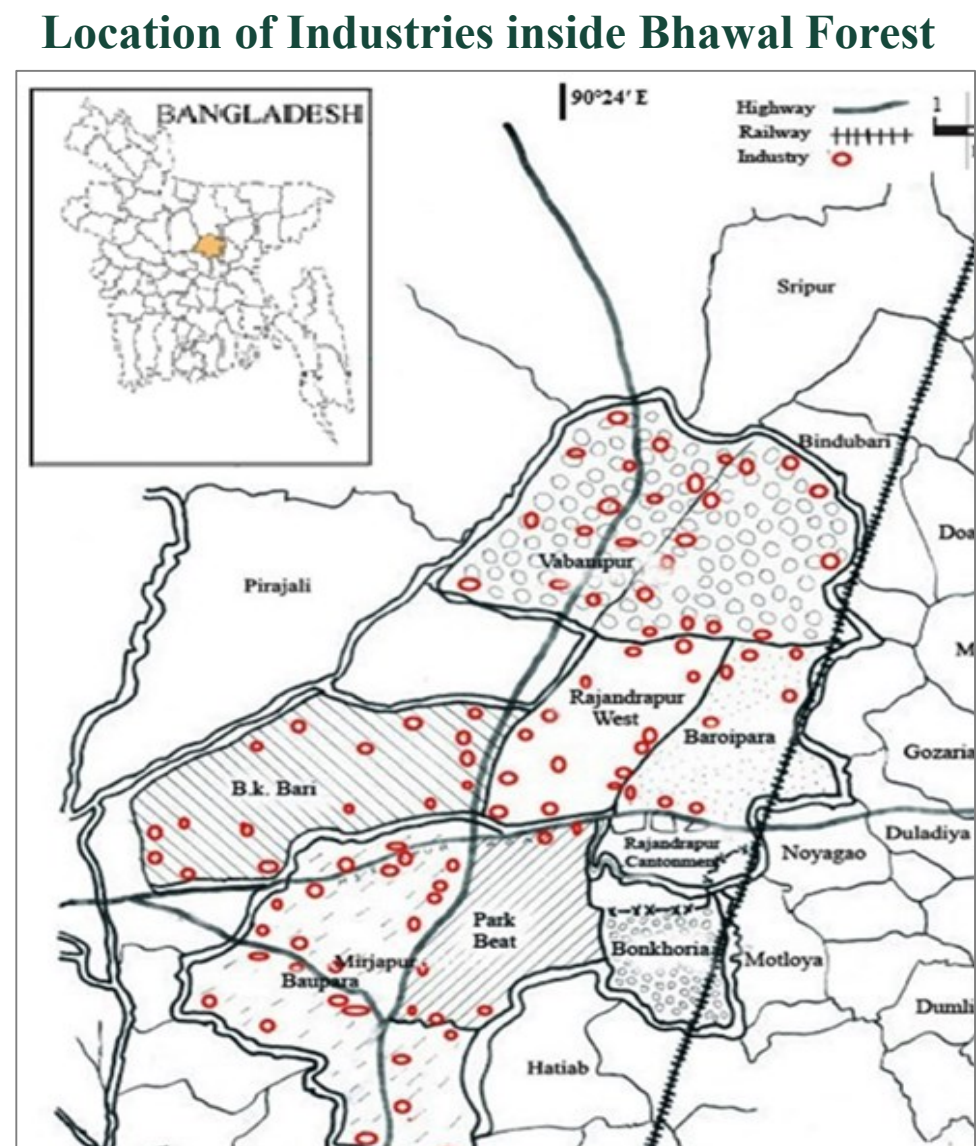
Source: Prepared by Author (2024) from KII and FGD

Encroached Land in BNP	Area (in Acre)
Total Encroached Land	989.36 (Land Value = 197872 crore BDT)
Recovered Land	326.36 (Only 30% of encroached land)
Industries	151.44
Cottage, Agriculture	40.43

Source: BFD, 2024



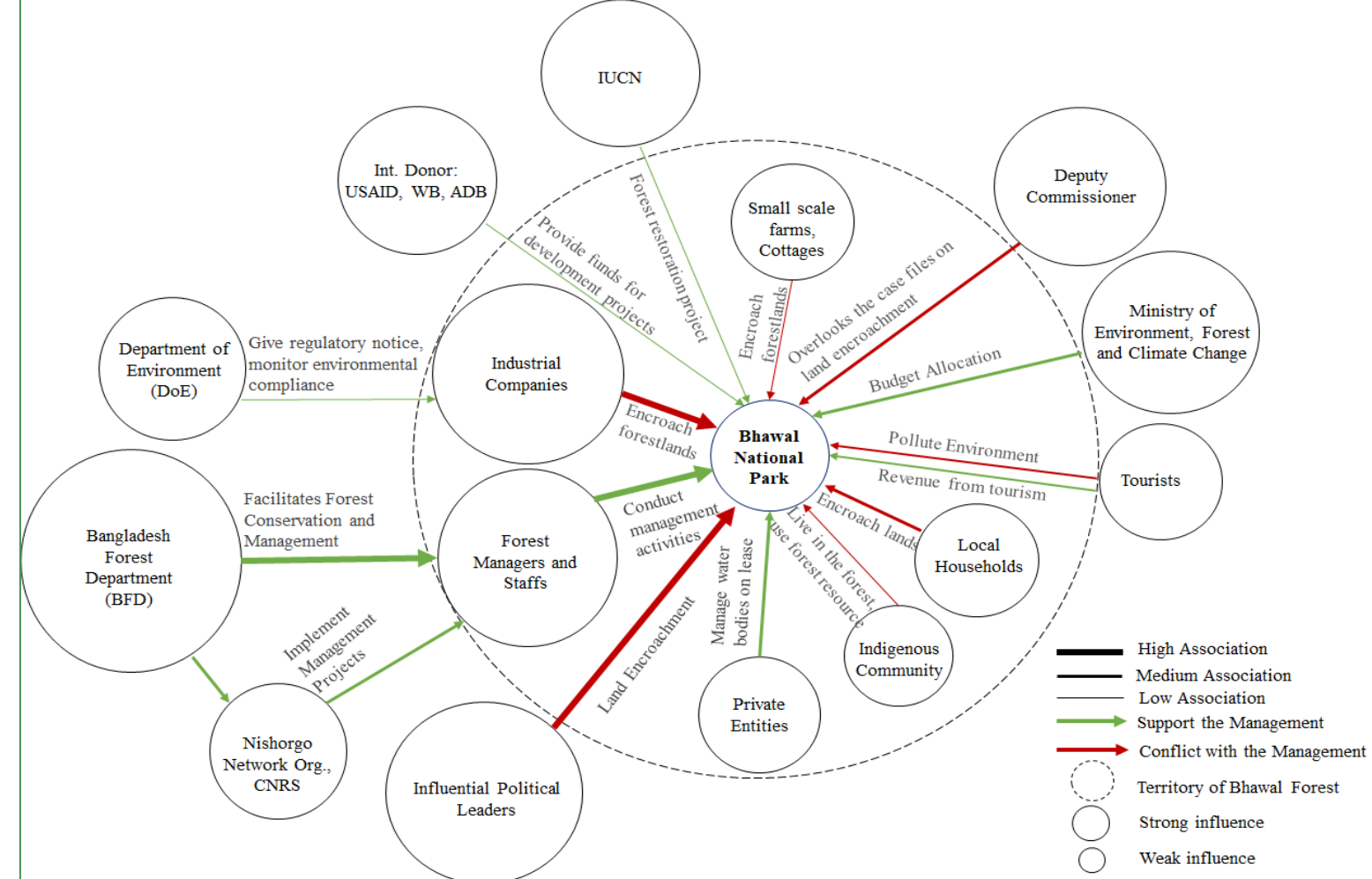
Source: BFD, 2024



Source: BFD, 2024

Management Effectiveness of Protected Areas

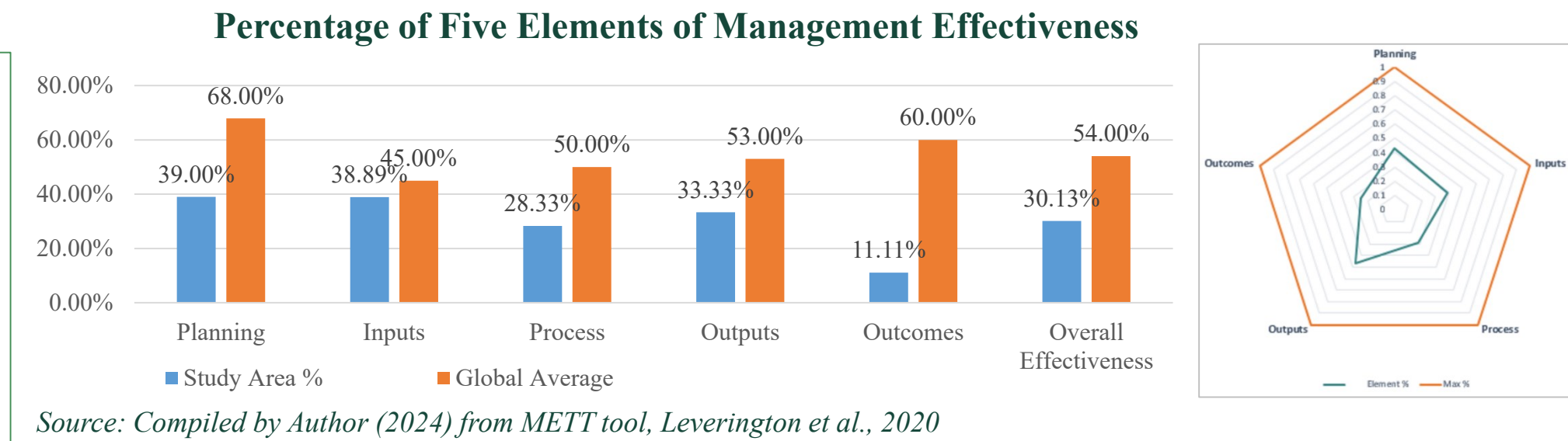
❑ Stakeholders Associated with the Management of Bhawal National Park



Source: Prepared by Author (2024) from KII and FGD

Current Scenario of Planning of Bhawal National Park	Level
Management plan has been prepared in 2014, but it did not progress into execution	Less Desirable
Involvement of all stakeholders is not ensured	Least Desirable
No regular work plan exists	Least Desirable
Insufficient number of people to manage the area	Less desirable

Source: Prepared by Author (2024) from KII



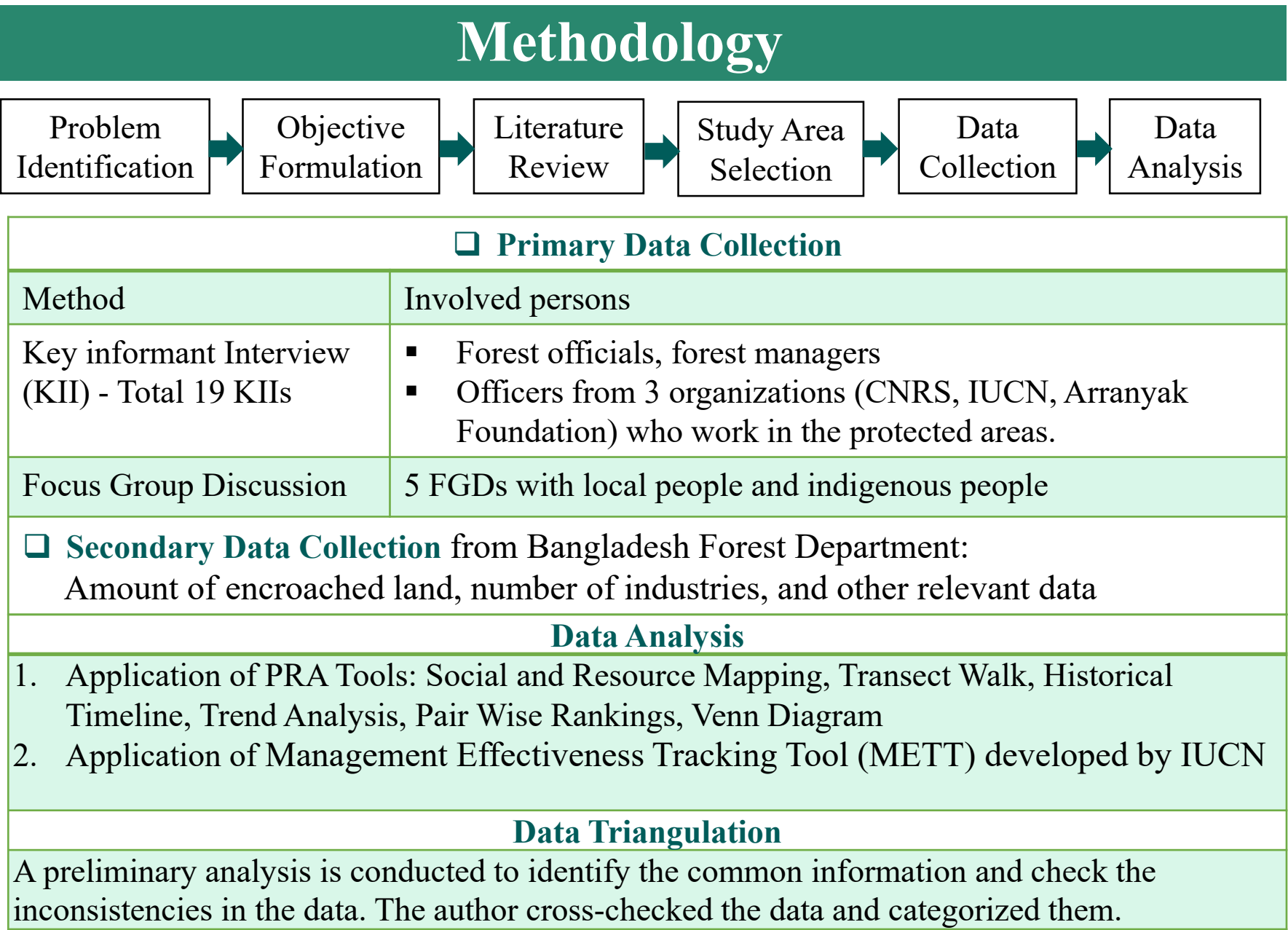
Source: Compiled by Author (2024) from METT tool, Leverington et al., 2020

❑ Management effectiveness of study area is only 30.13 percent. Significant drawbacks in the management of study area, reducing its conservation efforts to protect the forest ecosystem.

Current Scenario of Inputs of Bhawal National Park	Level
Insufficient number of people and equipment to manage the area	Less desirable
The available budget is inadequate for basic management	Less desirable
Current Scenario of Management Process of Bhawal National Park	
Even though there are some regulations , but there exists major weaknesses	Less desirable
The boundary of the protected area is not demarcated	Least desirable
Forest Managers have no enforcement capacity to enforce protected area legislation and regulations	Least desirable
Local communities are not involved in management decision	Least desirable

Source: Prepared by Author (2024) from KII

The protected area is unable to achieve its conservation objectives and provide desired outcomes. Lack of proper management approach and absence of coordination in the activities of different stakeholders weaken the forest management process of Bhawal National Park.



Major Findings

❑ 989.36 acres forestlands are illegally encroached by industries, agro based farms or private cottages. The total land valuation of the encroached lands is 197872 crore BDT.

❑ The Forest Department have filed 985 eviction cases against the illegal encroachers. Only 326.36 acres (30%) forestlands are restored from the encroachers.

❑ Before 2000, the issue of forest resource exploitation was extremely high as the forest resources were the only means of living for the poor people.

❑ Bangladesh Forest Department has weak institutional capacity to directly enforce the laws against the forest crimes.

❑ The current management process is excluding the local people from decision-making process and management activities.

Recommendation

❑ Addressing Inconsistent Policies and Plans: Forestry Master Plan aims to protect Sal forests, but existing land use plans, DAP (2022-2035) and UDP, 2024 of Gazipur do not conserve protected area.

❑ Law Enforcement Focus: Forest officials need direct enforcement power to combat illegal land encroachment.

❑ Central Committee for Monitoring and Management Plan Implementation : Establish a central committee led by Wildlife Management Division.

❑ Co-Management of the Study Area and Alternate Livelihood Opportunities for the Locals and Tribal Communities

Limitation

❑ Bangladesh Forest Department did not share some confidential data, for example details of land ownership in Bhawal forest.

❑ Forest managers were unavailable at times to conduct interviews.

❑ The industries could not be accessed for safety issues and could not conduct any interviews with industrialists.